First Non-Repeating Character

- **Find the first non-repeating character in a string using HashMap.**

```
package Hasp_map;
import java.util.HashMap;
import java.util.Scanner;
public class Non_repeat {
	public static void main(String[] args) {
		// TODO Auto-generated method stub
		Scanner sc = new Scanner(System.in);
    System.out.print("Enter a string: ");
    String input = sc.nextLine();

    HashMap<Character, Integer> charCountMap = new HashMap<>();

    for (int i = 0; i < input.length(); i++) {
      char ch = input.charAt(i);
      charCountMap.put(ch, charCountMap.getOrDefault(ch, 0) + 1);
    }

    boolean found = false;
    for (int i = 0; i < input.length(); i++) {
      char ch = input.charAt(i);
      if (charCountMap.get(ch) == 1) {
        System.out.println("First non-repeating character is: " + ch);
        found = true;
        break;
    }
    if (!found) {
      System.out.println("All characters are repeating.");
    }


	}
	}}
```

```
Enter a string: tapabrata
All characters are repeating.
All characters are repeating.
First non-repeating character is: p
```

Online Shopping Cart

- **Use List for cart, Map for product-price mapping.**

```
package Hasp_map;
import java.util.*;
public class Shopping_cart {
	public static void main(String[] args) {
		Map<String, Double> product = new HashMap<>();
		product.put("Laptop", 1200.00);
		product.put("Mouse", 25.50);
		product.put("Keyboard", 45.00);
```

```
                product.put("Monitor", 250.00);

                List<String> cart = new ArrayList<>();

                cart.add("Laptop");
                cart.add("Mouse");
                cart.add("Mouse");
                cart.add("Monitor");

                System.out.println("Items in cart: " + cart);

double total = 0.0;

                for (String item : cart) {
                        if (product.containsKey(item)) {
                                total += product.get(item);
                        } else {
                                System.out.println("Warning: " + item + " is not in the catalog.");
                        }
                }

                System.out.println("Total Order Cost: " + total);
        }
}
```

```
Items in cart: [Laptop, Mouse, Mouse, Monitor]
Total Order Cost: 1501.0
```

Library Management

- **Use Map<Book, Boolean> to track availability.**

```
package Hasp_map;

import java.util.HashMap;
import java.util.Map;

class Book {
   String title;

   Book(String title) {
      this.title = title;
   }

   public String toString() {
      return title;
   }
}

public class Library_manage {
```

```
    public static void main(String[] args) {
            Map<Book, Boolean> library = new HashMap<>();

Book b1 = new Book("Java Programming");
Book b2= new Book("Data Structures");
Book b3= new Book("Algorithm Design");

library.put(b1, true);
library.put(b2, true);
library.put(b3, false);

borrowBook(library, b1);
borrowBook(library, b3);


    }
    public static void borrowBook(Map<Book, Boolean> lib, Book book) {
if (!lib.containsKey(book)) {
    System.out.println("Book not found");
} else if (lib.get(book)) {

    lib.put(book, false);
    System.out.println("Success: borrowed '" + book + "'.");
} else {
    System.out.println(book + "' checked out.");
}

}}
```

```
Success: borrowed 'Java Programming'.
Algorithm Design' checked out.
```